CSC 251 HOMEWORK 9

DIGIT CLASSIFICATION USING SUPERVISED LEARNING

THE ASSIGNMENT

This assignment employs machine learning techniques to identify hand-written images of single digits using 3 different classifiers: Logistic Regression, k Nearest Neighbors, and Support Vector Machine classifiers. The accuracy of each will be computed and a sampling of images will be displayed.

BACKGROUND

The **Modified National Institute of Standards and Technology (**MNIST) **database** is a well-known dataset for computer vision applications that includes 70,000 hand-written digits images, 60,000 for training and 10, 000 for testing. To be specific, each image is a .jpg file containing a hand-written digit with a label 0 to 9. The images are gray scale with a size of 28x28 pixels. For example:



The **data.zip** file is provided for you. After unzipping the file, the data directory will include /train and /test subdirectories. Each subdirectory contains 10 subfolders with a folder name indicating the label of the images. You will use 3 different classifiers from the sklearn library that will be covered in class: logistic regression, k-nearest neighbor (kNN) and Support Vector Machine (SVM). For more information about these classifiers, you can refer to the following videos:

Logistic Regression: https://www.youtube.com/watch?v=YMJtsYIp4kg

K Nearest Neighbors (kNN): https://www.youtube.com/watch?v=0p0o5cmgLdE

Support Vector Machine (SVM): https://www.youtube.com/watch?v=05VABNfa1ds

PROCEDURE

1) Open the Jupyter notebook provided for this homework and fill in the cell with your name, date, and homework description at the top in a Markdown cell. If you have not already installed the `sklearn` library, do so by typing at the command line:

```
pip install scikit-learn
```

In the next cell, import all necessary libraries. The first one, `os` is provided for you. You'll also need:

a) To assist with loading the dataset:

```
import matplotlib.image as mpimg
```

b) For the Logistic Regression classifier:

```
from sklearn.linear_model import LogisticRegression
```

c) For the k Nearest Neighbors classifier:

```
from sklearn.neighbors import KNeighborsClassifier
```

d) For the SVM classifier:

```
from sklearn.svm import LinearSVC
```

e) You'll also want to include: `NumPy` and `matplotlib.pyplot`

2) Load the training and test datasets. The script for the `train_dataset` list is provided for you. Notice, the outer for loop creates a path to each subdirectory, then in the inner loop the images are loaded and appended to the `train_dataset` list as an (image, label) tuple element provided the images are the correct size, (28, 28). Write the script to follow the same procedure for the `test_dataset` list.

3) Using script, output the number of items in each list and the size of each image.

4) Convert the training images and test images into `NumPy` objects. The features (images) should be reshaped and appended to one ndarray, while the labels should be appended to another. Since the images are each 28 x 28 pixels, the final dimensions of each of the vectors should be 1 x 784, flattened into a single row.

Repeat this step for your test images.

5) Train the first classifier by fitting it with the training data. Then use the classifier to predict your test data images. Report the accuracy of the classifier on the test dataset in a comment. You might not have 100% success!

6) Visualize the results of your classifier on samples of your test dataset. Write a loop to iterate through your original test images and output only the results of every 500th image. You may find the following script helpful in displaying your image:

```
plt.imshow(image, cmap="gray")
```

To suppress the axis from being displayed:

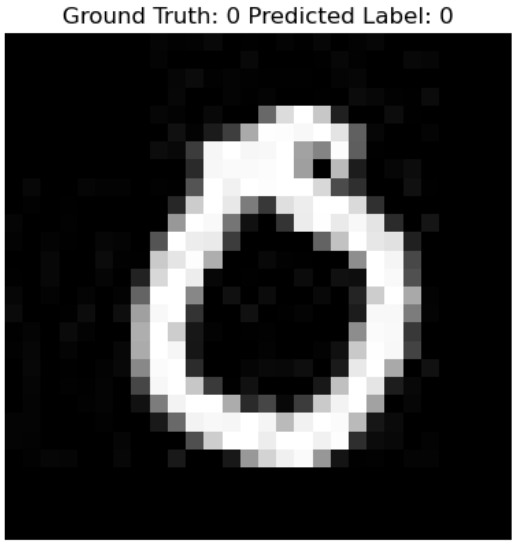
```
plt.axis('off')
```

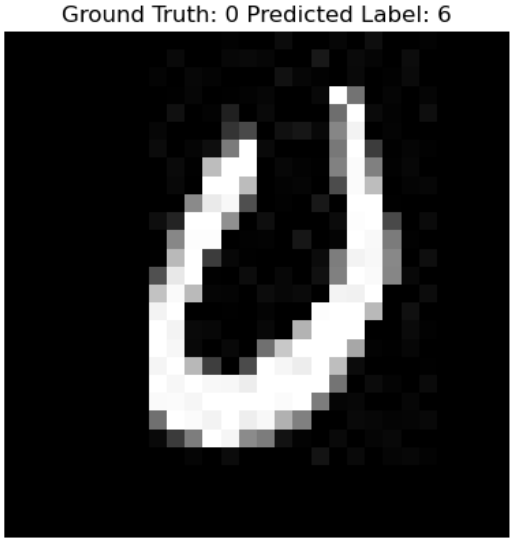
To display each selected image as you iterate, you'll want to include:

```
plt.show()
```

Your output should be labeled like the following. (When you create a figure title, remember that the label is stored in that tuple!) Note, some results will indicate successful classification while others may indicate failure

to classify properly.





7) Repeat for the other 2 classifiers. NOTE: ONLY RUN ONE CLASSIFIER AT A TIME! Comment out the script to run the others but include it in your notebook along with comment on its accuracy.

8) At the very bottom of your notebook, include your answers to the following questions in a comment:

a) Compare the performance (accuracy and runtime) of the 3 classifiers. Specify how your notebook was run: Google Colab, JupyterLite, or locally on laptop/desktop (specify OS/type)

b) Why do you think a specific classifier has better or poorer performance?

9) Save your notebook after running the Logistic Regression classifier and submit this file to Moodle. This will allow us to grade your notebook without taking the time (whew!) to run it. Points will be deducted if we need

to run your notebook and cannot view your results.

FILE SUBMISSION

Submit a SINGLE notebook called `HW9LastNameFirstInitial.ipynb` on Moodle, saved with the Logistic Regression classifier fit and images showing.